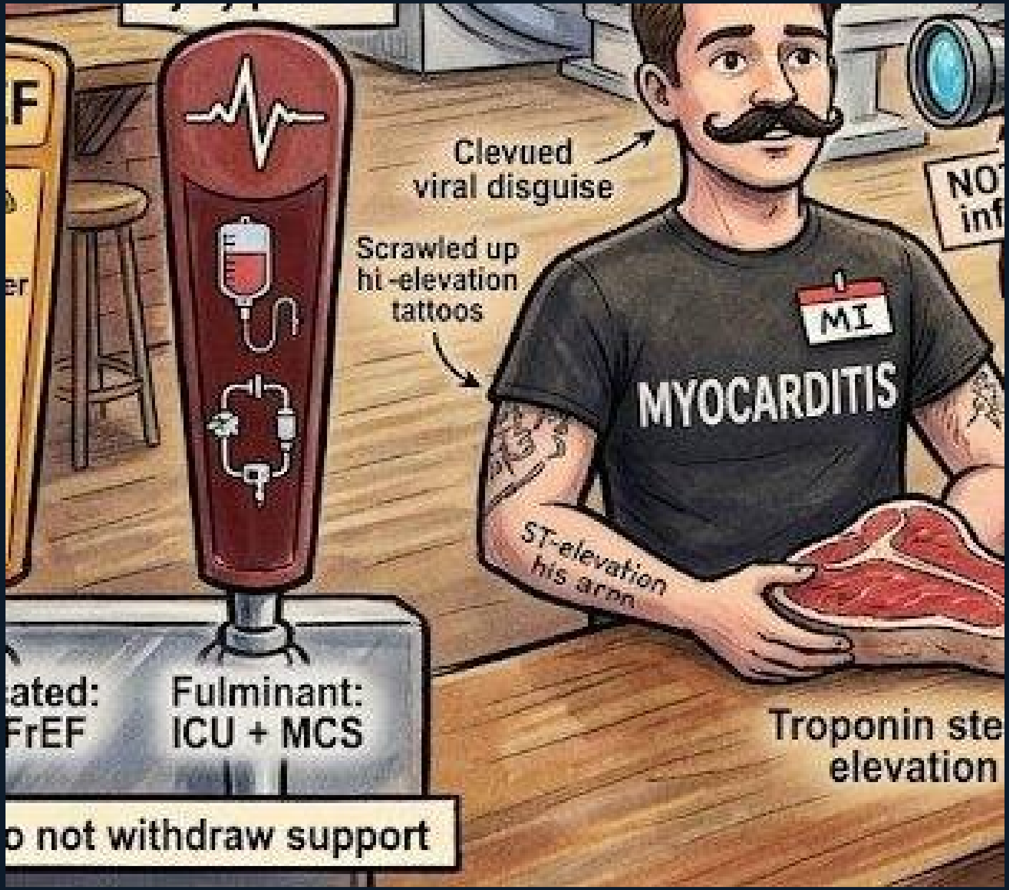
Scene C — Myocarditis: The MI Mimic

Craft Brewery Taproom · 7 board questions



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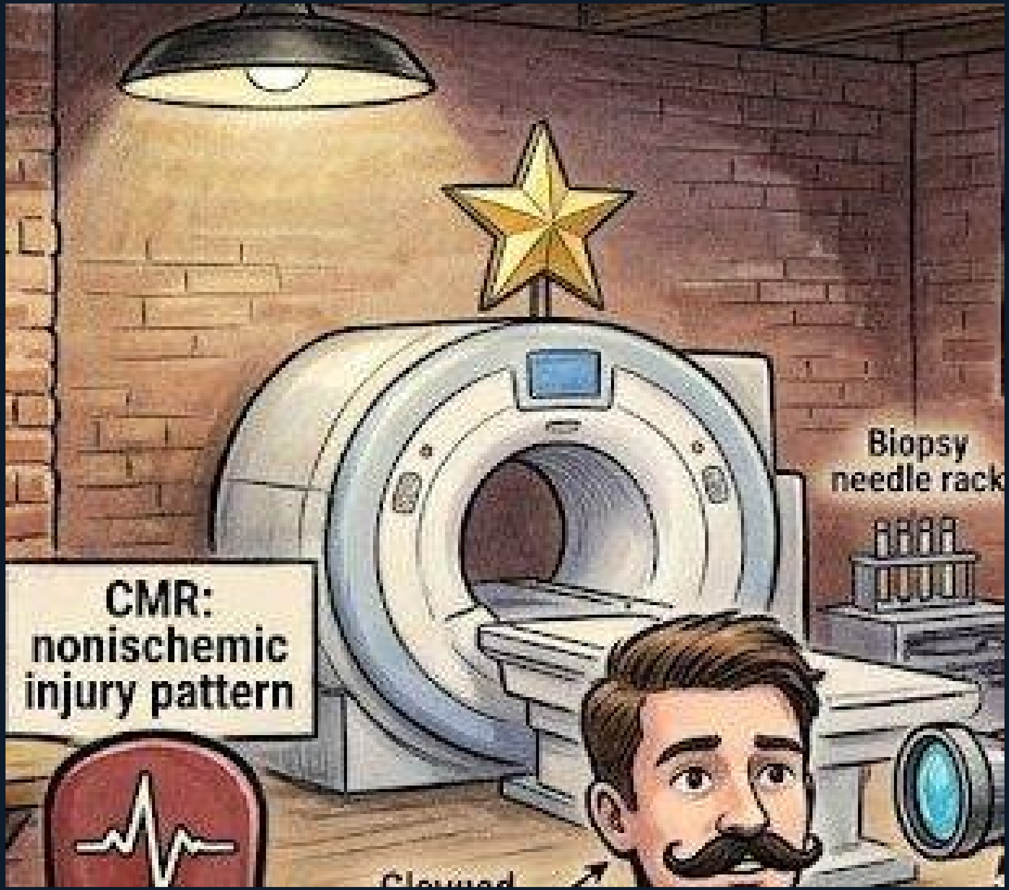
The MI Impersonator

The young man wears a fake mustache, an 'MI' nametag, but his shirt reads 'MYOCARDITIS.' T-bone steak = troponin. ST-elevation tattoos on his arm. This is the core trap: myocarditis presents with chest pain, ST elevation, and elevated troponin — identical to STEMI. The discriminating clue: young patient, viral illness days to weeks prior. If coronary angiogram shows normal vessels, think myocarditis, Takotsubo, or MINOCA. The board will present this scenario and ask next step — CMR, not thrombolytics.



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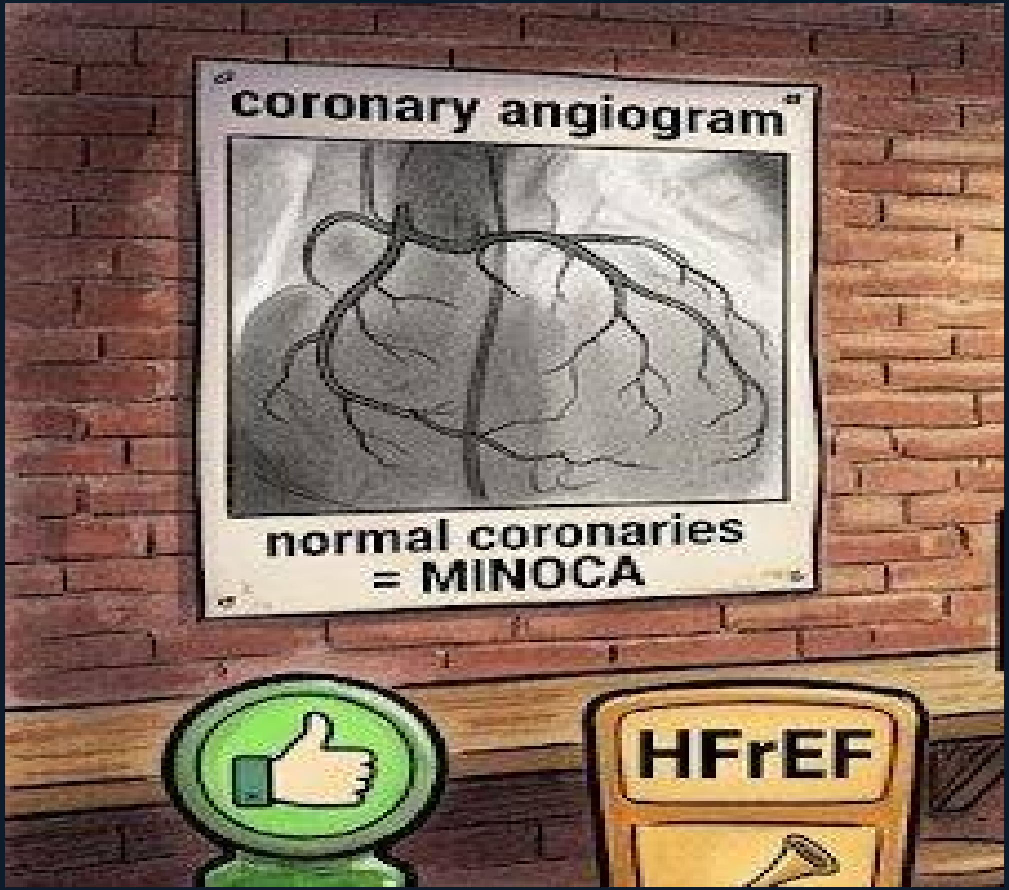
CMR: Gold Standard ★

The CMR machine gets the gold star. CMR shows myocardial edema on T2 and nonischemic late gadolinium enhancement — the injury pattern doesn't follow a coronary distribution (subepicardial or midmyocardial, not subendocardial as in infarct). Board rule: 'What is the best next test to confirm myocarditis?' CMR. Not biopsy, not troponin trend, not stress test. CMR also characterizes fibrosis for risk stratification. The correct diagnosis is established by: typical presentation + elevated troponin + CMR showing nonischemic injury pattern.



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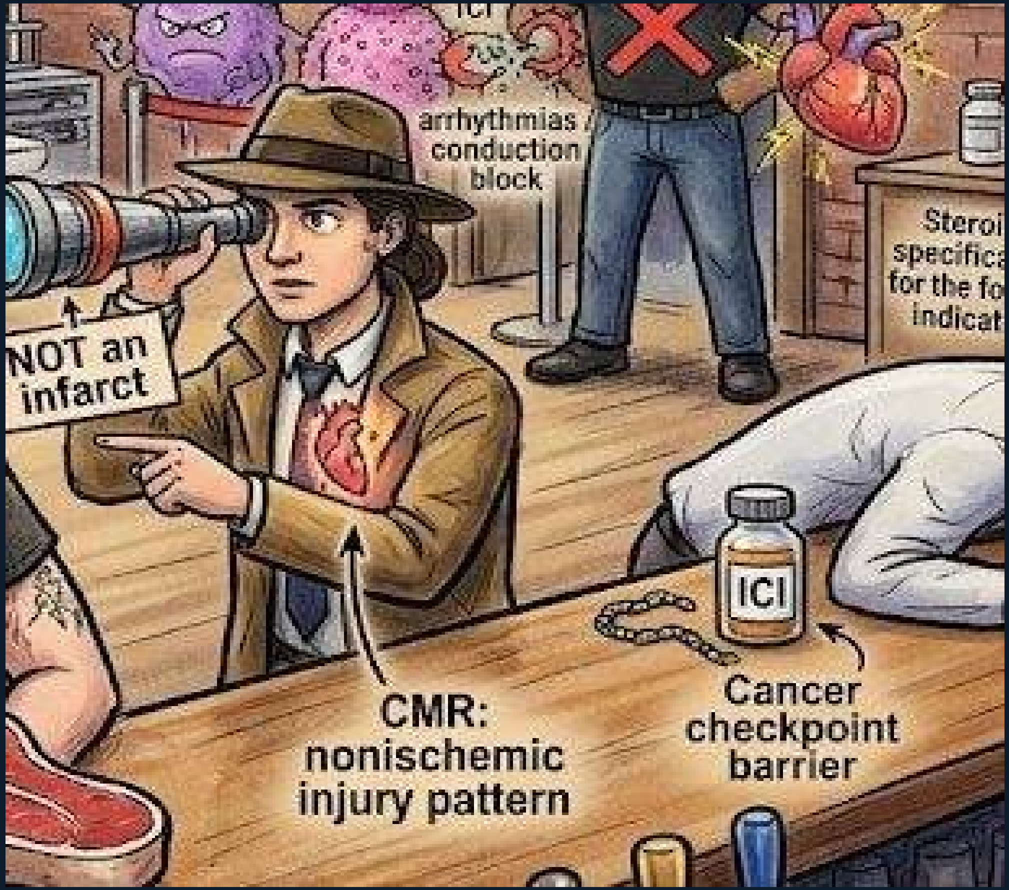
Normal Coronaries = MINOCA

The coronary angiogram shows clear vessels — MINOCA (Myocardial Infarction with Non-Obstructive Coronary Arteries). Myocarditis accounts for up to 1/3 of MINOCA cases. When ST elevation and troponin rise but catheterization shows clean coronaries, the differential includes myocarditis, Takotsubo, coronary spasm, and SCAD. Myocarditis is the most common in young patients with viral prodrome. The board tests this with: 'Catheterization shows normal coronaries — what is the most likely diagnosis in a 28-year-old post-viral illness?'



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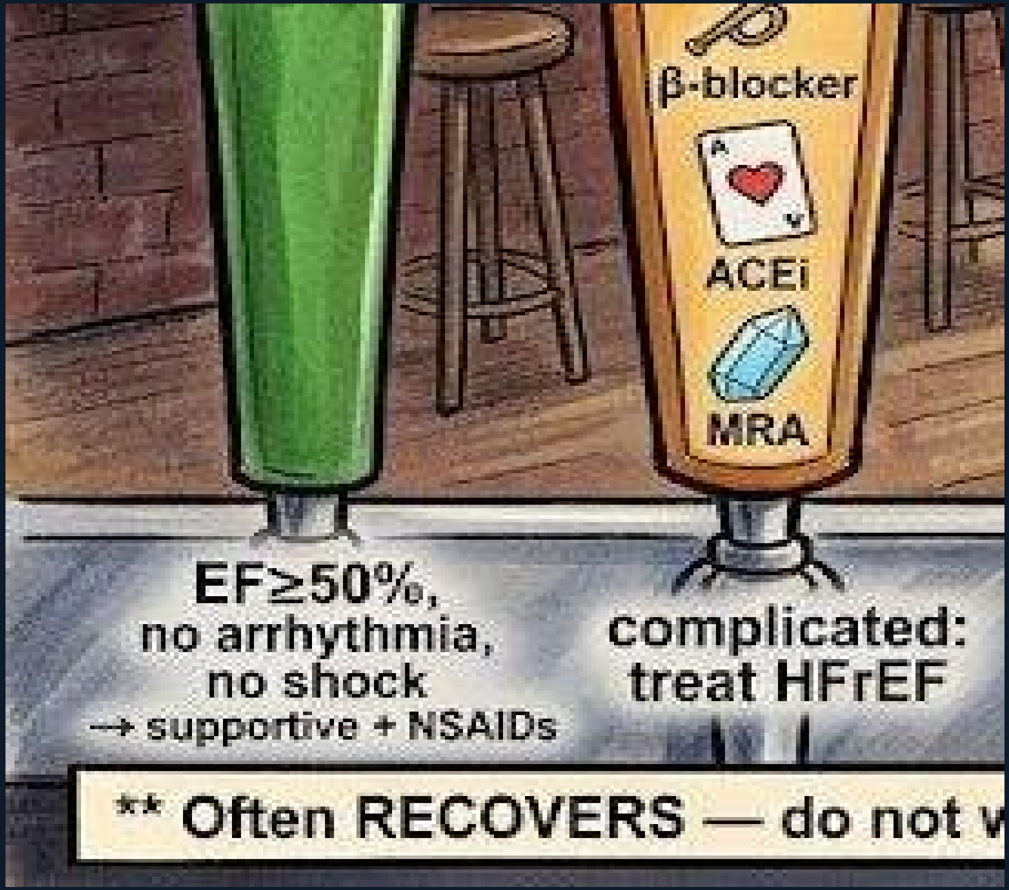
Biopsy Bouncer: 4 Allowed Types

The bouncer blocks routine biopsy but allows four types past. Giant cell myocarditis: aggressive, requires immunosuppression — must identify by biopsy. Eosinophilic myocarditis: responds to steroids. ICI myocarditis: cancer immunotherapy trigger, requires immediate high-dose steroids. Arrhythmias or conduction block: when CMR nondiagnostic and biopsy changes management. Board trap: 'Presumed viral myocarditis with NSVT — biopsy needed?' No. NSVT alone in typical viral myocarditis context does not mandate biopsy.



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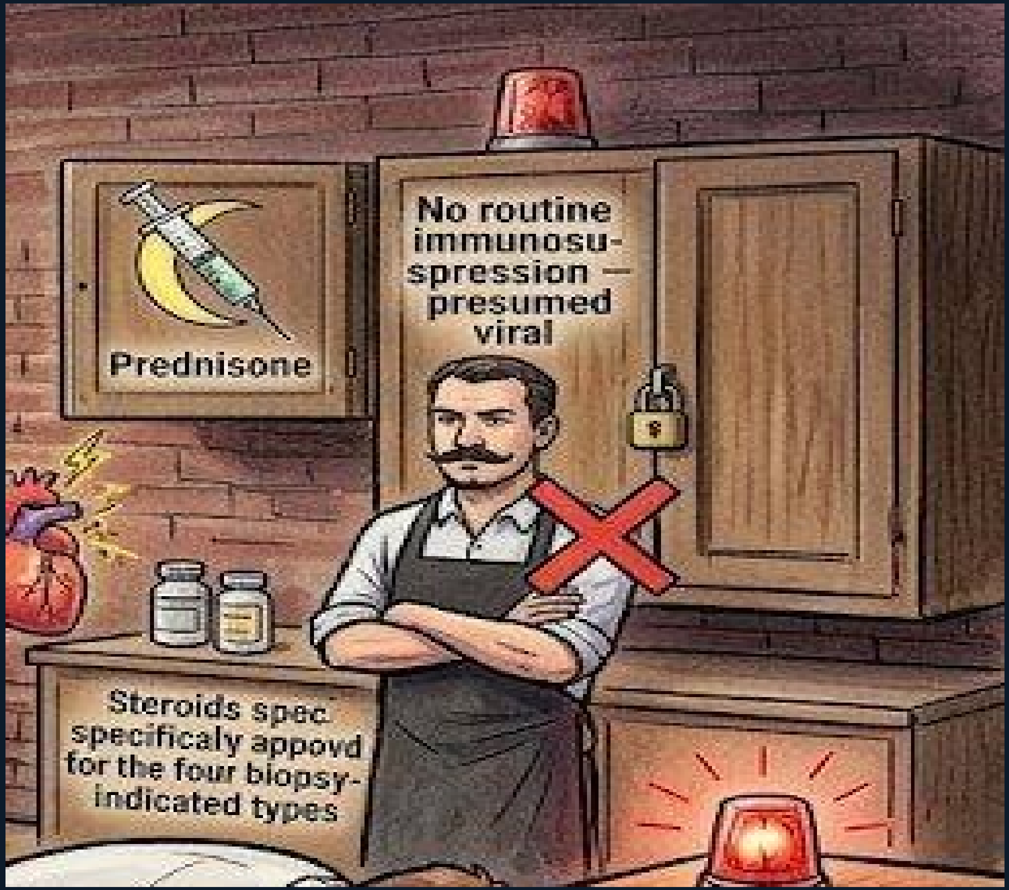
Three-Tap Severity System

Green tap: uncomplicated (EF ≥50%, no significant arrhythmias, no shock) → supportive + NSAIDs for pain. Excellent prognosis: 95% transplant-free survival at 5 years. Amber tap: complicated (EF <50%) → HFrEF GDMT (ACEi, beta-blocker, MRA). Red tap: fulminant (cardiogenic shock) → ICU, high-dose inotropes, mechanical circulatory support. Board pearl: 'Fulminant myocarditis — often RECOVERS if supported through the acute phase.' 50% recover to near-normal EF. Do not withdraw support prematurely.



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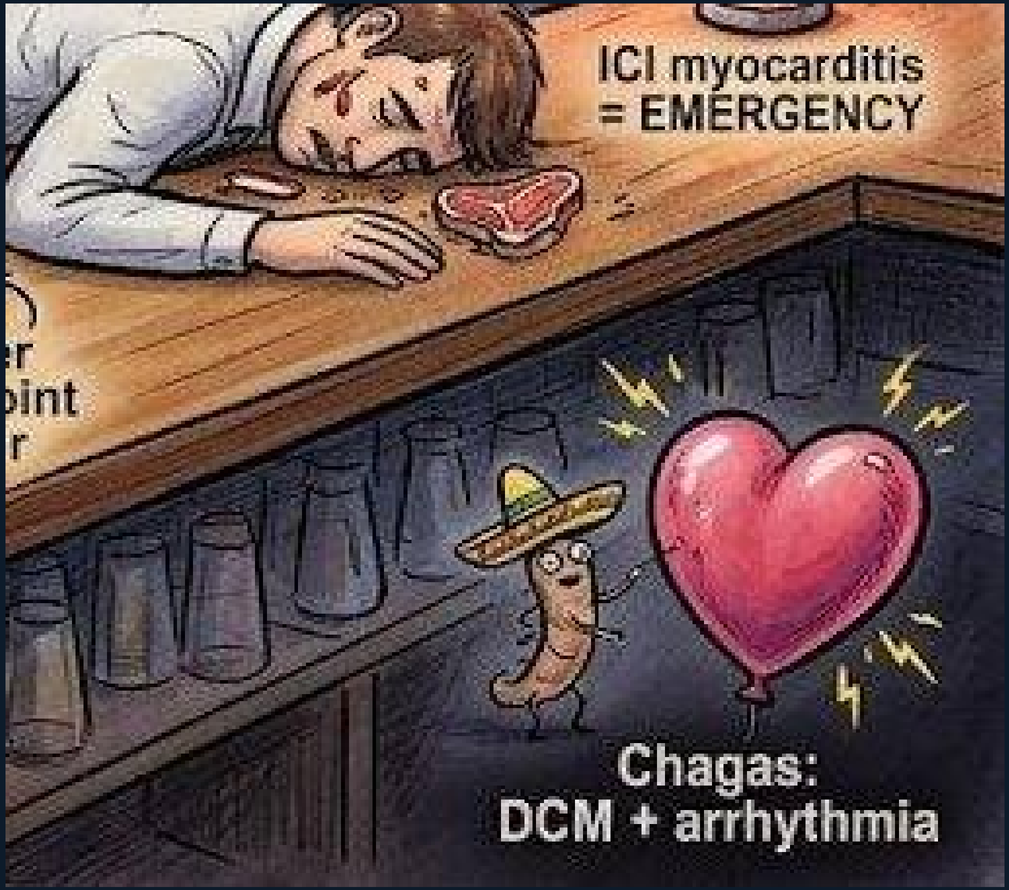
Steroids: Locked + ICI Emergency

Locked cabinet + red X: do NOT routinely immunosuppress presumed viral myocarditis. Steroids impair viral clearance and worsen outcomes. Small unlocked cabinet: steroids ARE appropriate for the four biopsy-confirmed types. ICI emergency red siren: cancer patient on checkpoint inhibitor develops troponin elevation with arrhythmia or heart block = ICI myocarditis = EMERGENCY. Stop immunotherapy immediately + high-dose corticosteroids. Mortality from ICI myocarditis is substantial — delay is fatal.



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Chagas + Viral Titers Useless

Chagas bug (Trypanosoma cruzi with sombrero): endemic in Latin America, causes dilated cardiomyopathy with prominent ventricular arrhythmias and RBBBlock pattern. Consider in any DCM patient from Latin America — it has a specific treatment protocol. Viral titers poster with red X: routine viral serology (Coxsackie titers, adenovirus titers) has no diagnostic value in acute myocarditis — low sensitivity, low specificity, does not change management. Do not send viral titers. Do CMR.

Scene C — Quick Reference

Myocarditis mimics STEMI

STE + troponin ≠ MI — viral prodrome is the clue

CMR = gold standard ★

Nonischemic LGE + edema = confirms diagnosis

MINOCA: 1/3 = myocarditis

Normal coronaries → consider myocarditis

Biopsy: 4 types only

Giant cell, eosinophilic, ICI, rapid progression

Uncomplicated: EF≥50% → supportive

95% 5-yr transplant-free survival

Fulminant → MCS — often recovers

50% recover near-normal EF if bridged

No routine steroids in viral

Worsens outcome — impairs viral clearance

ICI myocarditis = EMERGENCY

Stop ICI + high-dose steroids immediately

Chagas: DCM + arrhythmia

Latin America + RBBBlock + DCM → test for Chagas

Viral titers: NOT useful

No sensitivity/specificity — do not send